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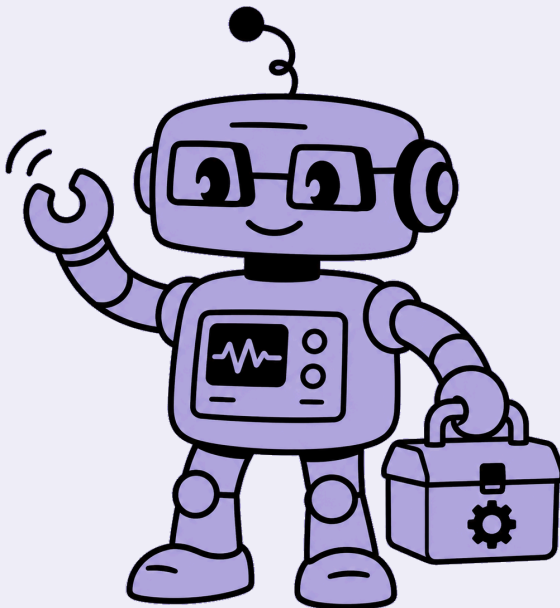
openhouse

robotics • level 2

◆ electronics ◆

your portfolio of the openhouse experience

ages 8-12



why electronics?

This is an **engineering** book. Engineers build real machines and work out how they work.

Inside almost every machine – a torch, a fan, a robot vacuum, an automatic door – is a **circuit**: electricity flowing round a loop of wires.

Electronics is how those circuits work. You'll build machines that light up, move, and even sense the world – and say why they work.

You grow four **skills** and build five real **machines**. You start from the very first circuit – no need to have built machines before. The next pages show the skills, the machines, the words, and how to stay safe.

the skills you build

You grow all four skills on **every machine**. Each grows in four **outcomes** – the machines climb them a little further each time. **presenting & explaining** is new this level.



observing & understanding

1. name each block's job – source, path, control, output →
 2. record a comparison – brighter, faster, which way →
 3. work out the rule – series vs parallel, the trigger point →
 4. **predict what a circuit will do, then prove it**
-



building & making

1. wire it from the card, checking each join →
 2. build it and read its schematic →
 3. build a circuit with several blocks →
 4. **design your own working circuit for a goal**
-



problem solving

1. notice the fault and predict where it is →
 2. check one block at a time →
 3. change one thing and record the effect →
 4. **debug step by step, and complete a missing wire**
-



presenting & explaining

1. name the circuit and its blocks →
 2. explain how it works, in the electronics words →
 3. explain why, using your result →
 4. **present it and take a question you didn't expect**
-

the machines

the machines you build

Five real machines in one book. Each teaches one big electronics idea – from a first circuit to a machine that senses and responds.

parking barrier

open & closed circuits

wind turbine

polarity

soccer bot

polarity reversal

cleaning bot

one battery, many jobs

sensor-controlled crane

input & output

the words you'll learn

Real engineers use real words. You'll learn the **blocks** (the parts you hold) and the **ideas** (what they do). Tick each one off as you learn it.

the blocks

power block – gives power

jumper wires – carry the electricity

led block – lights up

resistor block – keeps the led safe

switch block – turns it on and off

motor block – makes movement

direction switch – forward or backward

speed dial – changes the speed

sensor block – the robot's eyes

motor driver – like the brain

the ideas

circuit – the complete loop electricity travels round

open & closed – a gap stops it; joined, it works

polarity – electricity has a direction – + and –

series & parallel – share one path, or each its own path

input & output – what a machine senses, and what it does

staying safe with electricity

Electricity is useful, but it must be treated with care. Read these before you start.



The battery blocks in your kit are **safe** to touch and build with.



The power in a **wall socket** is much, much stronger. **Never** poke anything into a socket.



Never touch a switch, plug or wire with **wet hands**.



Wires wear a plastic coat – an **insulator** – to keep the electricity safely inside. Don't use a wire with the coat torn off.



If something smells hot or looks burnt, **stop** and tell an adult.

what's inside this book

For **every machine** you do six things – you can do them on your own:

1

explore

a story, a quick activity, the words, and a fun fact

2

experiment

run the real circuit, then answer a question about it

3

complete the circuit

finish a circuit – add the missing block or wire

4

build & name

build the circuit, then name its blocks

5

solve

fix a circuit that won't work – and stay safe

6

show

show it and say what you learned

At the end – the **level-up build**: a machine that senses and responds, the check you're ready for the next level.



meet the parking barrier

Aarav's building has a car-park gate that lifts when the guard presses a button. One evening it wouldn't open and cars piled up. Aarav spotted a loose wire, showed the guard, and together they fixed the loop – and the gate lifted again.

words to know

circuit**switch****current**

 **your turn:** tick what is a simple circuit:

- ☐ a torch
- ☐ a cushion
- ☐ a doorbell



did you know?

the flow of electricity going round and round the loop is called **current**.



parking barrier · experiment & observe



observing & understanding

outcome 1 – you're learning to: name each block's job – source, path, control, output



do the experiment

Build the loop, then trace it with your finger from the + end of the battery, through every block, back to the – end.

now work it out: How many blocks does the current flow through on its way round the loop?



circle your answer:



all of them, in one path



only some of them



draw the circuit as a schematic – use the block symbols



design a circuit

your turn to be the engineer. design a circuit that does a real job:

the brief

Design a circuit for a car-park gate: the guard can switch it ON and OFF, and a light shows when it is on.

remember: the light must only be on when the switch is on

1 List the blocks you will need:

my blocks ...

2 Draw your circuit as a schematic – then check current can flow all the way round.

 draw your design here



building & making

Build your parking barrier circuit from your blocks and circuit card.

Make it so it **lifts and drops when you flip the switch.**

☐

tick when it works

- 1 **Draw** your circuit in the box below.
- 2 **Label** these two blocks on your drawing:

your labels

the power block

the switch block

 draw your parking barrier circuit – then label its blocks



problem solving

outcome 1 – you're learning to: notice the fault and predict where it is

The parking barrier's light won't come on – one wire has come loose.
Find the break and make the loop work again.



Draw your fix – what did you change to make it work?

 draw your fix here



in real life

A torch, a doorbell and a table lamp are all simple circuits – a loop the electricity flows around.



presenting & explaining

show your parking barrier to a friend. say three things:

1. **what it is** – “this is my parking barrier. it works as a circuit with a switch.”
2. **its blocks** – point to the power block and the switch block.
3. **what it does** – “it lifts and drops when you flip the switch.”

 the new words i learned – write what each one means:

circuit

switch

current



meet the wind turbine

Meera visited a wind farm on a hill. Her cousin asked how the blades made a light glow. Meera built a little turbine – and when she swapped the wires by mistake, the blades spun backwards. They both laughed, then worked out why.

words to know

motor

led

polarity



your turn: tick what must go the right way round:

- ☐ a battery in a remote
- ☐ a picture book
- ☐ a torch battery



did you know?

a battery stores **charge** – like a store of energy, waiting to flow.



observing & understanding

outcome 2 – you're learning to: record a comparison – brighter, faster, which way



do the experiment

Wire the motor one way and note the spin direction. Now swap the two wires and note it again.

now work it out: Swapping the wires makes the motor spin:



circle your answer:



the opposite way



the same way



draw the circuit as a schematic – use the block symbols



design a circuit

your turn to be the engineer. design a circuit that does a real job:

the brief

Design a circuit where a switch lets you choose WHICH WAY the motor spins – forward or backward.

remember: swapping the direction must not need any rewiring

1 List the blocks you will need:

my blocks ...

2 Draw your circuit as a schematic – then check current can flow all the way round.

 draw your design here



building & making

Build your wind turbine circuit from your blocks and circuit card.

Make it so it **spins the blades and lights the led.**

☐

tick when it works

1 Draw your circuit in the box below.

2 Match each block to what it does – draw a line.

the motor block

lights up when power flows the right way

the led block

spins to turn the blades

 now draw your circuit



problem solving

outcome 2 – you're learning to: check one block at a time

The wind turbine's motor is spinning the wrong way. Change one thing so it spins the way you want.



Draw your fix – what did you change to make it work?

 draw your fix here



in real life

Batteries in a remote must go the right way round, or nothing works – that's polarity.



presenting & explaining

show your wind turbine to a friend. say three things:

1. **what it is** – “this is my wind turbine. it shows polarity.”
2. **its blocks** – point to the motor block and the led block.
3. **what it does** – “it spins the blades and lights the led.”

 the new words i learned – write what each one means:

motor

led

polarity



meet the soccer bot

The class built soccer bots for a friendly match. Sam's bot could only drive straight, so it kept missing the ball. A friend showed Sam the direction switch, and soon Sam's bot could turn and join in too.

words to know

direction switch

reverse

steer



your turn: tick what can drive forwards and backwards:

☐

a remote-control car

☐

a brick wall

☐

a toy robot



did you know?

electricity flows from the + side all the way round the loop to the – side.



observing & understanding

outcome 2 – you're learning to: record a comparison – brighter, faster, which way



do the experiment

Try every position of the direction switch, with two motors wired in.

now work it out: To make the bot TURN (not just go straight), the two motors must spin:



circle your answer:



opposite ways



the same way



draw the circuit as a schematic – use the block symbols



design a circuit

your turn to be the engineer. design a circuit that does a real job:

the brief

Design a circuit that drives TWO motors and can reverse both together, so the bot can go forward, back and turn.

remember: one control should flip both motors

1 List the blocks you will need:

my blocks ...

2 Draw your circuit as a schematic – then check current can flow all the way round.

 draw your design here



building & making

Build your soccer bot circuit from your blocks and circuit card. Make it so it **drives forwards, back, and turns.**

☐

tick when it works

- 1 **Draw** your circuit in the box below.
- 2 **Fill in each block** – write it from the word bank.

flips the motors forwards or back ...

the _____

drive and steer the bot ...

the _____

word bank

the direction switch

the two motors

 now draw your circuit



problem solving

outcome 2 – you're learning to: check one block at a time

The soccer bot only drives straight – it won't turn. Change one thing so it can steer.



Draw your fix – what did you change to make it work?

 draw your fix here



in real life

A remote-control car changes direction the same way – a switch flips the motor.



presenting & explaining

show your soccer bot to a friend. say three things:

1. **what it is** – “this is my soccer bot. it reverses its motors to steer.”
2. **its blocks** – point to the direction switch and the two motors.
3. **what it does** – “it drives forwards, back, and turns.”

 the new words i learned – write what each one means:

direction switch _____

reverse _____

steer _____



meet the cleaning bot

Priya's grandmother finds sweeping tiring. Priya built a cleaning bot that runs its wheels and brush together from one battery, to help her grandmother keep the floor clean.

words to know

series

parallel

speed



your turn: tick what runs many things from one battery:

☐

a robot vacuum

☐

a garden rock

☐

a mobile phone



did you know?

voltage is the push behind the flow – more push makes the motors spin faster.



observing & understanding

outcome 3 – you're learning to: work out the rule – series vs parallel, the trigger point



do the experiment

Run the two motors in series, then in parallel, and compare their speed.

now work it out: The motors run FASTER when they are wired in:



circle your answer:



parallel (each its own path)



series (one shared path)



draw the circuit as a schematic – use the block symbols



design a circuit

your turn to be the engineer. design a circuit that does a real job:



the brief

Design a circuit that runs **THREE** motors from **ONE** battery, each getting full power, with a single switch to turn it all off.

remember: full power to each motor means a parallel circuit

1 List the blocks you will need:

my blocks ...

2 Draw your circuit as a schematic – then check current can flow all the way round.



draw your design here



building & making

Build your cleaning bot circuit from your blocks and circuit card.

Make it so it **runs all its motors from one battery**.

☐

tick when it works

- 1 **Draw** your circuit in the box below.
- 2 **Label** these two blocks on your drawing:

your labels

the battery

the speed dial

 draw your cleaning bot circuit – then label its blocks



problem solving

outcome 3 – you're learning to: change one thing and record the effect

One of the cleaning bot's motors won't run; the others are fine. Find out why and fix it.



Draw your fix – what did you change to make it work?

 draw your fix here



in real life

A robot vacuum runs its wheels and brush from one battery, just like your cleaning bot.



presenting & explaining

show your cleaning bot to a friend. say three things:

1. **what it is** – “this is my cleaning bot. it runs many motors from one battery.”
2. **its blocks** – point to the battery and the speed dial.
3. **what it does** – “it runs all its motors from one battery.”

 the new words i learned – write what each one means:

series

parallel

speed



meet the sensor-controlled crane

At the port, huge cranes stop on their own if someone walks underneath. Reza built a small sensor crane that stops when a hand comes near – a safe crane, he told his little sister, so no one gets hurt.

words to know

sensor

input

output



your turn: tick what senses and responds on its own:

☐

an automatic tap

☐

a wooden chair

☐

a night-light



did you know?

a **sensor** turns what it sees into a tiny electric signal the driver can read.



observing & understanding

outcome 4 – you're learning to: predict what a circuit will do, then prove it



do the experiment

Move an object slowly toward the sensor and watch the motor. Note the distance where it reacts.

now work it out: The sensor makes the motor react when the object is:



circle your answer:

☐

close enough (the trigger point)

☐

any distance at all



draw the circuit as a schematic – use the block symbols



design a circuit

your turn to be the engineer. design a circuit that does a real job:

the brief

Design a circuit where a **SENSOR** stops the motor by itself when something comes near.

remember: show the path: input → driver → output

1 List the blocks you will need:

my blocks ...

2 Draw your circuit as a schematic – then check current can flow all the way round.

 draw your design here



building & making

Build your sensor-controlled crane circuit from your blocks and circuit card. Make it so it **stops on its own when your hand is near**.

☐

tick when it works

1 Draw your circuit in the box below.

2 Match each block to what it does – draw a line.

the sensor

the brain – it switches the motor

the motor driver

the eyes – it senses your hand

 now draw your circuit



problem solving

outcome 4 – you're learning to: debug step by step, and complete a missing wire

The crane doesn't stop when your hand is near. Change one thing so the sensor makes it stop.



Draw your fix – what did you change to make it work?

 draw your fix here



in real life

Automatic taps, doors and night-lights all sense and respond – input to output.



presenting & explaining

show your sensor-controlled crane to a friend. say three things:

1. **what it is** – “this is my sensor-controlled crane. it senses and responds.”
2. **its blocks** – point to the sensor and the motor driver.
3. **what it does** – “it stops on its own when your hand is near.”

 the new words i learned – write what each one means:

sensor

input

output



your level-up build

This is your **level check**. Build **one machine that senses and responds** – a circuit that reacts to a sensor, or your own circuit that does a real job. Show all four skills on it.

✓ **you are ready for the next level when you can:**

- ☐ predict what a circuit will do, then prove it
- ☐ design your own working circuit for a goal
- ☐ debug step by step, and complete a missing wire
- ☐ present it and take a question you didn't expect

draw your level-up build here